

CO4 AT2

INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

Given Conditions

- Accuracy must be 90% or above.
- Latency must be 70 ms or below.
- The vehicle supports moderate computational resources.

Comparison of Methods

Method	Accuracy	Latency	Computation	Meets All Constraints?
Viola-Jones	82%	30 ms	Low	No
YOLO	93%	60 ms	Moderate	Yes
Deep Learning	97%	90 ms	High	No

Evaluation

Viola-Jones: Viola-Jones has a low latency of 30 ms and requires low computation. However, its accuracy is only 82%, which is below the required 90%. Therefore, it cannot be selected.

YOLO: YOLO provides 93% accuracy and 60 ms latency. Both values satisfy the required limits. It also requires moderate computation, which matches the available hardware resources.

Deep Learning: Deep Learning provides the highest accuracy of 97%, but its latency is 90 ms, which exceeds the maximum acceptable latency of 70 ms. It also requires high computational resources. Therefore, it does not satisfy the constraints.

Final Decision

YOLO is the most suitable method.

- Accuracy = 93% $\geq$ 90%
- Latency = 60 ms $\leq$ 70 ms
- Computation = Moderate

Therefore, YOLO provides the best balance of accuracy, low latency, and computational requirements for the autonomous vehicle vision system.

2. Smart Factory Product Inspection Selection

Given Conditions

- Accuracy must be 95% or above.
- Processing time must be 50 ms or below.
- Hardware supports moderate computation.

Comparison of Methods

Method	Accuracy	Processing Time	Computation	Meets All Constraints?
Traditional Image Analysis	91%	25 ms	Low	No
YOLO	96%	45 ms	Moderate	Yes
Deep Learning	98%	75 ms	High	No

Evaluation

Traditional Image Analysis: Traditional Image Analysis has a fast-processing time of 25 ms and requires low computation. However, its accuracy is only 91%, which is below the required 95%. Therefore, it may miss defective products and cannot be selected.

YOLO: YOLO provides 96% accuracy with a processing time of 45 ms. Both values satisfy the required limits. It also requires moderate computation, which is suitable for the available hardware.

Deep Learning: Deep Learning provides the highest accuracy of 98%, but its processing time is 75 ms, exceeding the maximum allowed 50 ms. It also requires high computational resources. Therefore, it does not satisfy the production-line requirements.

Final Decision

YOLO is the most suitable method.

- Accuracy = 96% $\geq$ 95%
- Processing time = 45 ms $\leq$ 50 ms
- Computation = Moderate

Therefore, YOLO provides the best balance between defect detection accuracy, processing speed, and available computational resources for the smart factory.